

Project Topic Assignment

FPGA Market Maker Accelerator

Description:

This project aims to design and implement a low-latency market making system using a DE1-SoC FPGA board as the primary engine. The FPGA will execute core components of a market making algorithm while the HPS handles network communication. Communication between the HPS and FPGA will be achieved through memory-mapped I/O using AXI bridges and on-chip RAM, allowing market data to be written directly into FPGA-accessible memory for processing.

The motivation for this project is to explore hardware acceleration techniques commonly used in high-frequency trading (HFT) systems, where latency and parallel computation provide significant advantages over CPU-only implementations. Over the course of two semesters, this project aims to create a prototype capable of receiving real-time market data, processing it on the FPGA, and returning trading decisions to a Linux-based application for execution in a paper-trading environment.

Background Research Needed:

First, it is necessary to understand the interconnection method between the HPS and FPGA Fabric. Secondly, it is essential to master the basic deployment and debugging process of embedded Linux, as well as network programming methods for subscribing to and parsing real-time market data via WebSocket. Simultaneously, it is crucial to research and implement a reliable shared memory data protocol and consistency guarantee mechanism between the HPS and FPGA. Finally, the market-making strategy logic needs to be simplified and engineered, extracting the computational content and input/output formats suitable for hardware implementation, ensuring its stable operation and functional demonstration.

Initial Projection and Final Outcome:

The initial goal is to establish network connectivity between the HPS and Linux, and a shared RAM communication path between the HPS and FPGA. This will ensure that when the HPS writes fixed test data, the FPGA can reliably read and verify it. Subsequently, WebSocket market data will be integrated, with key fields written to shared memory, and real-time calculations and simplified market-making strategies implemented on the FPGA. The final outcome will be an HPS system that obtains real-time market data from WebSocket and writes it to shared RAM, while the FPGA reads the data, performs low-latency calculations, and outputs market-making trigger signals.